

CS 224n Assignment 2: word2vec

Written Solutions

1. Understanding word2vec

(a) Cross-entropy form of the naive-softmax loss

Because $\mathbf{y}$ is one-hot with $y_o = 1$ and $y_w = 0$ for $w \neq o$,

$$-\sum_{w \in \mathcal{V}} y_w \log \hat{y}_w = -\log \hat{y}_o = -\log P(O = o \mid C = c) = J_{\text{naive-softmax}}(\mathbf{v}_c, o, U).$$

(b) Gradient with respect to the center vector

Let $\mathbf{z} = U^\top \mathbf{v}_c$ and $\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$. The softmax cross-entropy derivative is $\partial J / \partial \mathbf{z} = \hat{\mathbf{y}} - \mathbf{y}$. Since $\partial \mathbf{z} / \partial \mathbf{v}_c = U^\top$,

$$\boxed{\frac{\partial J_{\text{naive-softmax}}}{\partial \mathbf{v}_c} = U(\hat{\mathbf{y}} - \mathbf{y}).}$$

(c) Gradients with respect to the outside vectors

For each vocabulary word w ,

$$\frac{\partial J_{\text{naive-softmax}}}{\partial \mathbf{u}_w} = (\hat{y}_w - y_w) \mathbf{v}_c.$$

Equivalently,

$$\boxed{\frac{\partial J}{\partial \mathbf{u}_w} = \begin{cases} (\hat{y}_o - 1) \mathbf{v}_c, & w = o, \\ \hat{y}_w \mathbf{v}_c, & w \neq o. \end{cases}}$$

(d) Derivative of the sigmoid

For every component x_i of a vector $\mathbf{x}$,

$$\boxed{\frac{\partial \sigma(x_i)}{\partial x_i} = \sigma(x_i)(1 - \sigma(x_i)).}$$

Thus the elementwise derivative is $\sigma(\mathbf{x}) \odot (\mathbf{1} - \sigma(\mathbf{x}))$; the full Jacobian is the diagonal matrix containing these values.

(e) Negative sampling gradients

Write $a_o = \mathbf{u}_o^\top \mathbf{v}_c$ and $a_k = \mathbf{u}_k^\top \mathbf{v}_c$. Using $\frac{d}{dx}[-\log \sigma(x)] = \sigma(x) - 1$ and $\frac{d}{dx}[-\log \sigma(-x)] = \sigma(x)$ gives

$$\frac{\partial J_{\text{neg-sample}}}{\partial \mathbf{v}_c} = (\sigma(a_o) - 1)\mathbf{u}_o + \sum_{k=1}^K \sigma(a_k)\mathbf{u}_k,$$

$$\frac{\partial J_{\text{neg-sample}}}{\partial \mathbf{u}_o} = (\sigma(a_o) - 1)\mathbf{v}_c, \quad \frac{\partial J_{\text{neg-sample}}}{\partial \mathbf{u}_k} = \sigma(a_k)\mathbf{v}_c.$$

If a negative word is sampled more than once, its contributions are added. Negative sampling is more efficient because it updates only the true outside word and K sampled words, rather than computing and updating scores for the entire vocabulary.

(f) Skip-gram gradients over a context window

Let $S = \{j : -m \leq j \leq m, j \neq 0\}$. Since the skip-gram objective is a sum over outside words,

$$\frac{\partial J_{\text{skip-gram}}}{\partial U} = \sum_{j \in S} \frac{\partial J(\mathbf{v}_c, w_{t+j}, U)}{\partial U},$$

$$\frac{\partial J_{\text{skip-gram}}}{\partial \mathbf{v}_c} = \sum_{j \in S} \frac{\partial J(\mathbf{v}_c, w_{t+j}, U)}{\partial \mathbf{v}_c},$$

and, for every center-word vector $\mathbf{v}_w$ with $w \neq c$,

$$\frac{\partial J_{\text{skip-gram}}}{\partial \mathbf{v}_w} = \mathbf{0}.$$

2(c). Trained word vectors

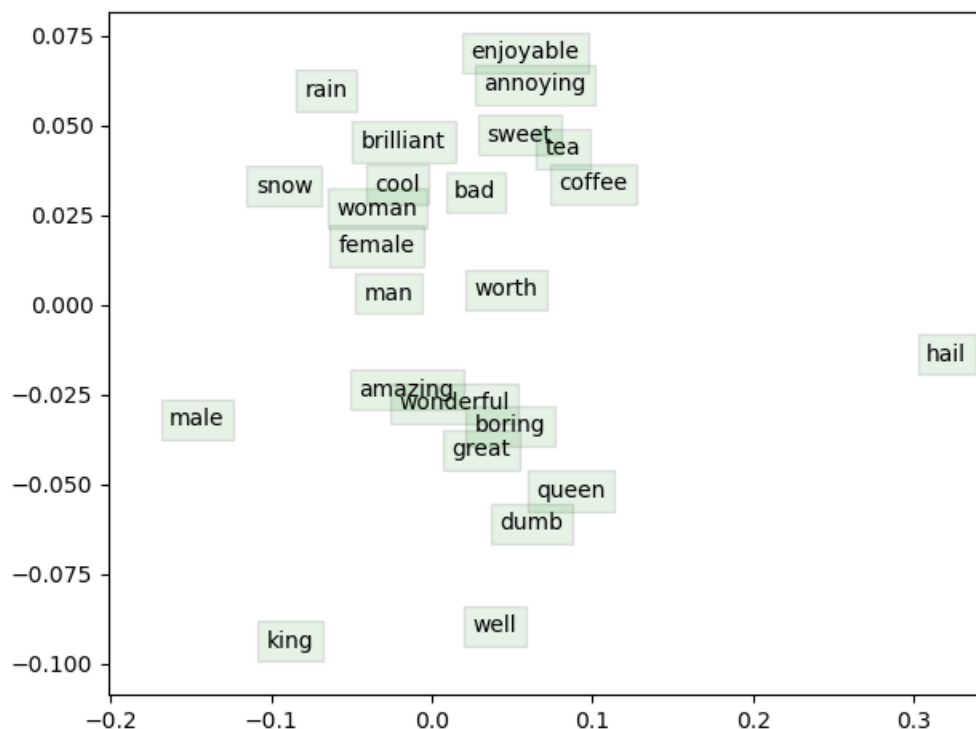


Figure 1: Two-dimensional projection of the word vectors learned from the Stanford Sentiment Treebank.

The projection shows meaningful local relationships: “coffee” is close to “tea,” “woman” is close to “female” and “man,” and “great,” “wonderful,” and “amazing” form a compact group. Some expected relationships are imperfect: “rain” is nearer to “snow” than to “hail,” while sentiment words such as “enjoyable”/“annoying” and “wonderful”/“boring” are not cleanly separated. This noise is reasonable because the vectors are trained on a modest corpus and a two-dimensional projection cannot preserve every relationship in the original ten-dimensional space.